

E.SANJAY

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Location : Thanjavur, Tamil Nadu 614626, India

GitHub: <https://github.com/Sanjay-E22> | Portfolio: <https://sanjay-e22.github.io/>

SUMMARY

I'm a passionate and curious data science and machine learning learner with strong skills in Python, Pandas, NumPy, and Scikit-learn. I enjoy working with data-cleaning it, analyzing it, and building models that solve real-world problems. I have hands-on experience in Natural Language Processing (NLP) projects and love uncovering insights from text data. I'm also comfortable working with Excel and Power BI to create dashboards and reports, and I use MySQL for managing and querying structured data. Always eager to learn new tools and techniques, I'm looking for opportunities where I can apply my skills, grow as a data professional, and contribute to impactful projects.

EDUCATION

Oct 2021 - May 2025

Bachelor of Engineering (B.E.) – Computer Science and Engineering

St. Joseph's College of Engineering & Technology, A.S Nagar, Elupatti, Thanjavur – 613 403, Tamil Nadu

CGPA: 8.3 / 10

Consistently strong in academics with a current CGPA of 8.3 and no history of arrears. Actively involved in academic projects, teamwork, and continuous learning. Known for taking initiative, being a dependable contributor, and always eager to learn and apply new skills in real-world scenarios.

SKILLS

Programming: Python, MySQL

Data Analysis: Pandas, NumPy, Excel (Advanced), Power BI

Visualization: Matplotlib, Seaborn, Excel Charts, Power BI

Machine Learning: Scikit-learn, Supervised Learning, Unsupervised Learning, Hyperparameter Tuning

NLP: NLTK, TF-IDF, Bag of Words, Word2Vec, Named Entity Recognition, Sentiment Analysis

Tools: Jupyter Notebook, Google Colab, VS Code, Git, GitHub, Streamlit

Soft Skills: Problem Solving, Time Management, Teamwork, Communication

CERTIFICATIONS

- NPTEL – Computer Networking and Internet Protocol
- NPTEL – Ethical Hacking
- Introduction to Machine Learning – Kaggle
- Data Analysis with Python – freeCodeCamp
- Basics of Python – SoloLearn
- Python Developer – SoloLearn

PROJECTS

1. YOLO-Based Fall Detection System

- Developed a real-time fall detection system using YOLOv5 and OpenCV to recognize and track human posture from video streams.
- Achieved ~90% precision in detecting fall vs. non-fall events using custom-trained deep learning model.
- Applicable for elder care safety, with potential use in hospitals and smart homes.

Tools: Python, OpenCV, YOLOv5, Deep Learning

Key Features: Real-time object detection, human safety monitoring, model trained on fall datasets.

2. Cancer Detection Web App

- Built a Logistic Regression model to predict cancer presence using patient medical features.
- Achieved 92% accuracy on test dataset; deployed model as an interactive Streamlit web app.
- Reduced diagnostic prediction time from minutes to milliseconds with instant predictions and user-friendly interface.

Tools: Python, Pandas, Scikit-learn, Streamlit

Key Features: Model training, prediction interface, user-friendly dashboard.

3. BERT-Based Resume Matcher

- Designed a BERT-based NLP system to match resumes and job descriptions using semantic similarity.
- Enabled automatic candidate shortlisting, improving HR efficiency by 30–40% in mock evaluations.
- Built and deployed an interactive web app using HuggingFace Transformers + Streamlit.

Tools: Python, BERT (Transformers), NLP, Streamlit

Key Features: Semantic similarity scoring, resume-job match ranking, interactive web UI.

4. Intrusion Detection System (IDS) using Random Forest

- Developed an Intrusion Detection System (IDS) to classify network traffic as normal or malicious.
- Achieved 95% precision and recall, optimized using GridSearchCV for parameter tuning.
- Processed >100,000 rows of security data; applicable for real-time cybersecurity threat detection.

Tools: Python, Scikit-learn, GridSearchCV, Pandas

Key Features: Classification of normal vs. malicious traffic, model optimization, evaluation metrics.

5. Rain Forecast Prediction System

- Built a rainfall prediction model using Decision Tree and Random Forest algorithms.
- Achieved 89% classification accuracy using historical weather datasets.
- Compared models with confusion matrix, ROC curves, and accuracy metrics to select best-performing one.

Tools: Python, Scikit-learn, Pandas, Matplotlib

Key Features: Data preprocessing, model comparison, rainfall classification.

INTERNSHIPS

Python Programming Trainee – WB – Jan 2024

Thanjavur, Tamil Nadu

- Completed intensive hands-on training in Python programming fundamentals, including variables, data types, loops, functions, and error handling.
- Applied Python automation techniques for file handling, string processing, and real-world tasks.
- Built multiple mini projects and logical programs to strengthen coding proficiency.
- Gained foundational skills essential for data science and scripting workflows.

Web Development Intern – IGNIZEE – Jun 2023 to Jul 2023

Chennai, Tamil Nadu

- Designed and developed responsive web pages using HTML, CSS, and JavaScript, focusing on layout improvements.
- Collaborated with front-end developers to improve UI/UX components based on project requirements.
- Participated in real-world client projects, learning about version control, code testing, and agile workflows.
- Strengthened front-end development skills critical for ML web app deployment (e.g., via Streamlit).

PORTFOLIO & CODE LINKS

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